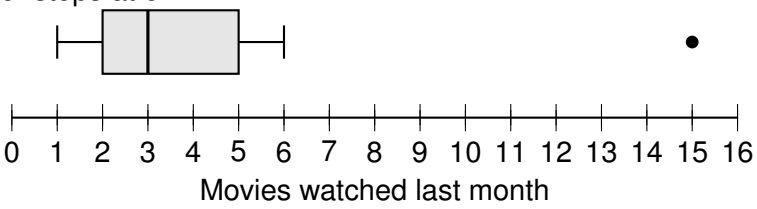
Unit 1 Review KEY

Name _____

One data set, every tool. 11 students said how many movies they watched last month.

3, 1, 15, 2, 4, 2, 6, 1, 3, 5, 2

1.1 Show the data. A dot plot puts one dot on the number line for each value.	 Most students watched 1 to 6 movies. One student watched 15.
1.2 Find the center. Σ means add up. n is the number of values. $\bar{x}$ is the mean.	$\bar{x} = \frac{\Sigma x}{n} = \frac{44}{11} = 4$ movies In order: 1, 1, 2, 2, 2, $\boxed{3}$, 3, 4, 5, 6, 15. Median = 3 movies The tail is on the right. It pulls the mean up, so mean $>$ median. The shape is **skewed right**.
1.3 Find the spread. Odd number of values? Leave the median out of both halves.	Lower half 1, 1, $\boxed{2}$, 2, 2. Q1 = 2 Upper half 3, 4, $\boxed{5}$, 6, 15. Q3 = 5 Five-number summary: 1, 2, 3, 5, 15 IQR = $Q_3 - Q_1 = 5 - 2 = 3$
1.4 Check for outliers. A value below the lower fence or above the upper fence is an outlier.	Lower fence = $Q_1 - 1.5 \times$ IQR = $2 - 4.5 = -2.5$ Upper fence = $Q_3 + 1.5 \times$ IQR = $5 + 4.5 = 9.5$ 15 is above 9.5, so **15 is an outlier**. On a box plot it is a dot. The whisker stops at 6. 

Read every box plot the same way. Each of the 4 parts holds about 25% of the data. A long part means the values are spread out, not that there are more of them.

Work with a partner. Compare answers when you finish each problem.

1.1 - Frequency Tables, Dot Plots, and Histograms

1. **I make and read a frequency table and a dot plot.**

20 students said how many times they went to the school library this month.

2, 0, 1, 3, 1, 2, 2, 0, 4, 1, 2, 3, 1, 2, 5, 1, 2, 0, 3, 2

(a) Make a frequency table.

Visits	Tally	Frequency
0	III	3
1	IIIII	5
2	IIIIIII	7
3	III	3
4	I	1
5	I	1

(b) Make a dot plot.



(c) What number of visits is most common? 2 visits (7 students)

(d) How many students went 3 or more times? 3 + 1 + 1 = 5 students

(e) How many students went fewer than 2 times? 3 + 5 = 8 students

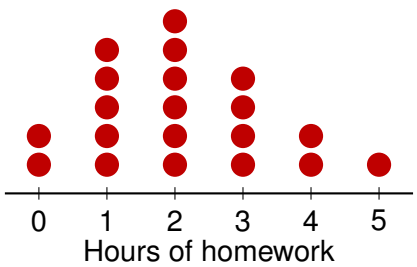
This frequency table shows how many hours some students spent on homework last night.

Hours	Frequency
0	2
1	5
2	6
3	4
4	2
5	1

(f) How many students are in the data? 2 + 5 + 6 + 4 + 2 + 1 = 20

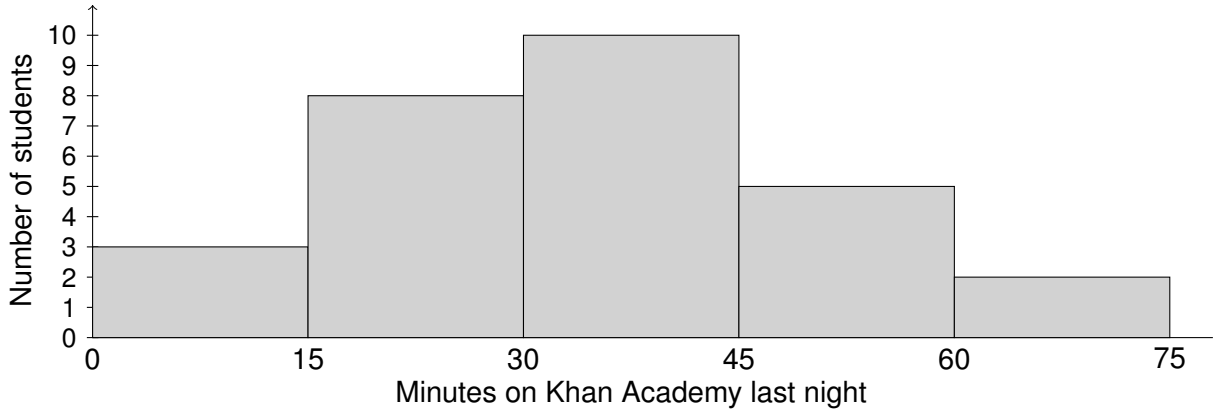
(g) How many students spent 3 or more hours? 4 + 2 + 1 = 7

(h) Make a dot plot of the same data.



2. **I make and read a histogram.**

This histogram shows how many minutes some students spent on Khan Academy last night.



(a) How many students spent from 15 to 30 minutes? 8

(b) How many students spent 45 minutes or more? $5 + 2 = 7$

(c) How many students are in the data? $3 + 8 + 10 + 5 + 2 = 28$

(d) One student spent exactly 30 minutes. Which bar is she in?
The 30 to 45 bar. A value on the line goes in the bar on the right.

(e) Can you tell how many students spent exactly 20 minutes? Explain.
No. The graph only shows that 8 students spent from 15 to 30 minutes. It hides the exact values.

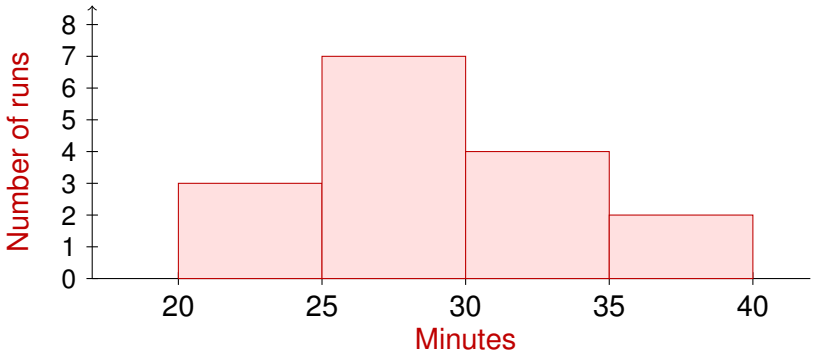
A runner recorded her time, in minutes, for 16 runs.

22, 25, 31, 28, 24, 35, 27, 29, 21, 33, 26, 30, 38, 25, 29, 32

(f) Make a frequency table. Remember the histogram rule.

Minutes	Frequency
20 to 25	3 (21, 22, 24)
25 to 30	7 (25, 25, 26, 27, 28, 29, 29)
30 to 35	4 (30, 31, 32, 33)
35 to 40	2 (35, 38)

(g) Make a histogram. Label both axes.


Number of runs
8
7
6
5
4
3
2
1
0
20
25
30
35
40
Minutes

1.2 - Mean, Median, and Shape

3. **I find the mean and the median of a data set.**

The temperature at noon for one week, in °F:

$$68,\ 72,\ 75,\ 70,\ 66,\ 72,\ 88$$

(a) Find Σx and $\bar{x}$.

$\Sigma x = 511$, $n = 7$, $\bar{x} = \dfrac{511}{7} = 73°\text{F}$

(b) Find the median.

In order: 66, 68, 70, $\boxed{72}$, 72, 75, 88. Median = 72°F

Scores on a 20-point quiz:

$$15,\ 18,\ 12,\ 20,\ 17,\ 14,\ 19,\ 9$$

(c) Find $\bar{x}$.

$\Sigma x = 124$, $n = 8$, $\bar{x} = \dfrac{124}{8} = 15.5$ points

(d) Find the median.

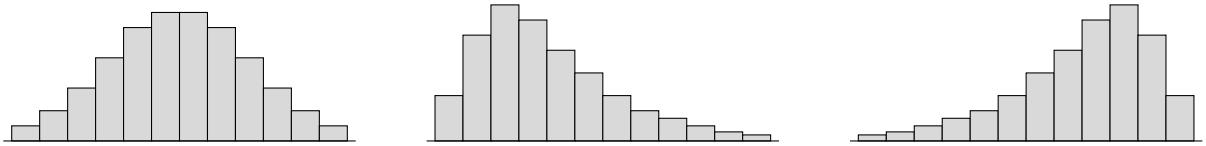
In order: 9, 12, 14, $\boxed{15}$, $\boxed{17}$, 18, 19, 20. Median = $(15 + 17) \div 2 = 16$ points

(e) A data set has $\Sigma x = 96$ and $n = 8$. Find $\bar{x}$. $\bar{x} = \dfrac{96}{8} = 12$

(f) Ana's first four quiz scores are 80, 75, 90, and 85. What score does she need on the fifth quiz so that the mean of all five scores is 84?

All five must add to $\Sigma x = 5 \times 84 = 420$. The first four add to $80 + 75 + 90 + 85 = 330$.
$420 - 330 = 90$. She needs a **90**.

4. **I tell the shape of a graph, and I choose the best measure of center.**



A **B** **C**

(a) For each histogram, name the shape. Then write $<$, $>$, or $\approx$ to compare the mean and the median.

Graph	Shape	mean ? median
A	symmetric	mean $\approx$ median
B	skewed right	mean $>$ median
C	skewed left	mean $<$ median

Seven students timed their trip to school, in minutes.

12, 10, 60, 14, 12, 17, 15

(b) Find the mean and the median.

$\Sigma x = 140$, $\bar{x} = \frac{140}{7} = 20$ minutes. In order: 10, 12, 12, $\boxed{14}$, 15, 17, 60. Median = 14 minutes

(c) Which better describes a typical trip, the mean or the median? Explain.

The median. 60 is far from the rest. It pulls the mean up. 6 of the 7 students take less than the mean.

Guess the shape of each data set: skewed left, symmetric, or skewed right. Explain.

(d) The length of phone calls. Most calls are short. A few calls are very long.

Skewed right. The few long calls make a tail on the right.

(e) The heights of all adult men in Missouri.

Symmetric. Most men are near the middle height. Fewer are very short or very tall.

(f) Scores on a test where most students did very well.

Skewed left. A few low scores make a tail on the left.

1.3 - Quartiles, IQR, and Box Plots

5. **I find the five-number summary and the IQR.**

Find the five-number summary and the IQR of each data set.

(a) 7, 15, 3, 12, 9, 20, 5, 11, 14

In order: 3, 5, 7, 9, 11, 12, 14, 15, 20. Lower half 3, 5, 7, 9: Q1 = 6. Upper half 12, 14, 15, 20: Q3 = 14.5

Min 3 Q1 6 Median 11 Q3 14.5 Max 20 IQR 8.5

(b) 41, 35, 50, 28, 46, 39, 44, 31, 52, 37, 48, 42, 33

In order: 28, 31, 33, 35, 37, 39, 41, 42, 44, 46, 48, 50, 52.
Lower half 28, 31, 33, 35, 37, 39: Q1 = (33 + 35) ÷ 2 = 34. Upper half 42, 44, 46, 48, 50, 52: Q3 = (46 + 48) ÷ 2 = 47

Min 28 Q1 34 Median 41 Q3 47 Max 52 IQR 13

(c) 6, 14, 9, 21, 11, 8, 17, 12, 25, 10

In order: 6, 8, 9, 10, 11, 12, 14, 17, 21, 25. Median = (11 + 12) ÷ 2 = 11.5
Lower half 6, 8, 9, 10, 11: Q1 = 9. Upper half 12, 14, 17, 21, 25: Q3 = 17

Min 6 Q1 9 Median 11.5 Q3 17 Max 25 IQR 8

6. **I make a box plot.**

12 students scored these points on a 50-point project.

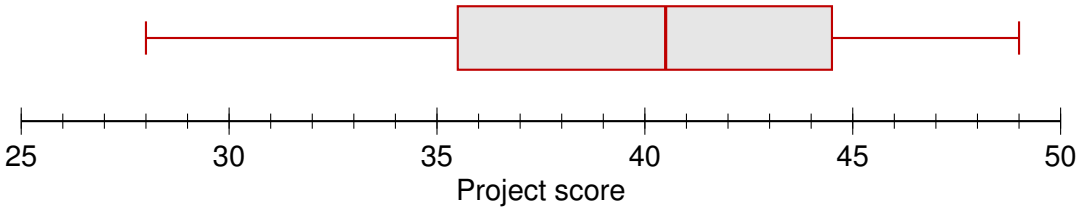
38, 42, 30, 45, 47, 35, 40, 44, 28, 49, 41, 36

(a) Find the five-number summary.

In order: 28, 30, 35, 36, 38, 40, 41, 42, 44, 45, 47, 49. Median = (40 + 41) ÷ 2 = 40.5
Lower half 28, 30, 35, 36, 38, 40: Q1 = (35 + 36) ÷ 2 = 35.5
Upper half 41, 42, 44, 45, 47, 49: Q3 = (44 + 45) ÷ 2 = 44.5

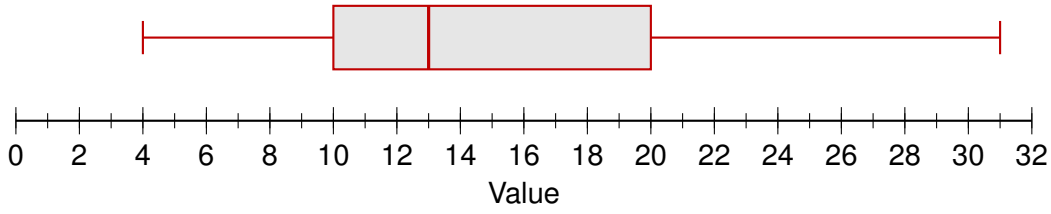
Min 28 Q1 35.5 Median 40.5 Q3 44.5 Max 49

(b) Make a box plot.



The five-number summary of a data set is 4, 10, 13, 20, 31.

(c) Make a box plot.



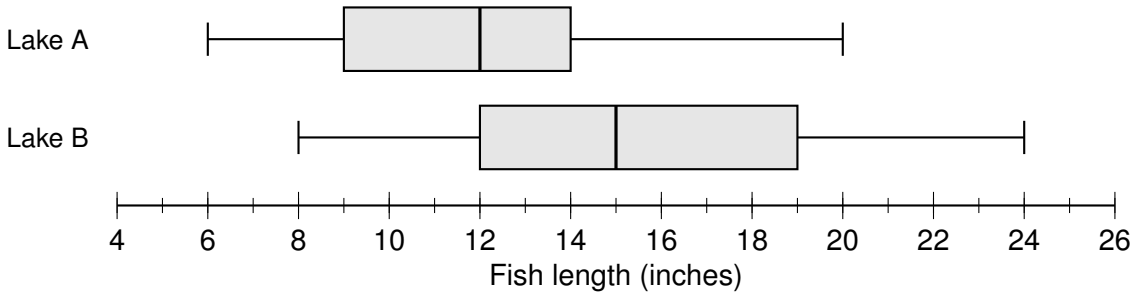
(d) Which part of the box plot is the longest? What does that tell you about the data?

The right whisker, from 20 to 31. The top 25% of the values are spread out.

1.4 - Reading Box Plots and Outliers

7. **I read and compare box plots.**

These box plots show the lengths of fish caught in two lakes.



(a) Write the five-number summary for Lake A. 6, 9, 12, 14, 20

(b) Which lake has the larger median? Lake B (15 inches is more than 12 inches)

(c) Find the IQR of each lake. Lake A: $14 - 9 = 5$ inches. Lake B: $19 - 12 = 7$ inches.

(d) Which lake's fish lengths are more consistent? Explain.
Lake A. Its IQR (5) and its range (14) are both smaller than Lake B's (7 and 16).

(e) About what percent of the fish in Lake B are longer than 12 inches? 75%

(f) About what percent of the fish in Lake A are from 9 to 14 inches long? 50%

(g) True or false? Explain. *Lake B's right whisker is the longest part, so most of Lake B's fish are from 19 to 24 inches long.*
False. Each part holds about 25% of the fish. A long part means those lengths are spread out.

8. **I use the 1.5 × IQR rule to find outliers.**

11 students scored these points on a test.

65, 30, 72, 61, 68, 78, 63, 66, 60, 69, 64

(a) Find Q1, Q3, and the IQR.
In order: 30, 60, 61, 63, 64, $\boxed{65}$, 66, 68, 69, 72, 78. Lower half 30, 60, 61, 63, 64: Q1 = 61
Upper half 66, 68, 69, 72, 78: Q3 = 69. IQR = 69 − 61 = 8

(b) Find the lower fence and the upper fence.
1.5 × 8 = 12. Lower fence = 61 − 12 = 49. Upper fence = 69 + 12 = 81

(c) Which values are outliers? Explain.
30 is an outlier, because it is below 49. 78 is **not** an outlier, because it is not above 81.

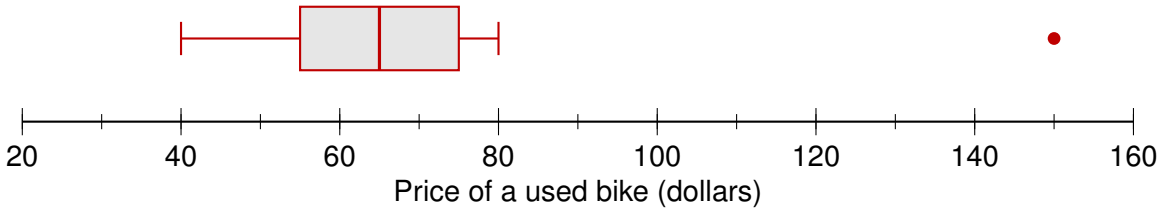
A store sells 8 used bikes. Here are the prices, in dollars.

66, 40, 150, 60, 80, 50, 70, 64

(d) Find Q1, the median, Q3, and the IQR.
In order: 40, 50, 60, $\boxed{64}$, $\boxed{66}$, 70, 80, 150. Median = (64 + 66) ÷ 2 = 65
Lower half 40, 50, 60, 64: Q1 = (50 + 60) ÷ 2 = 55. Upper half 66, 70, 80, 150: Q3 = (70 + 80) ÷ 2 = 75
IQR = 75 − 55 = 20

(e) Find the fences. Which values are outliers?
1.5 × 20 = 30. 55 − 30 = 25 and 75 + 30 = 105. 150 is an outlier.

(f) Make a box plot. Show each outlier as a dot.



(g) Why does the right whisker stop at 80?
80 is the largest price that is not an outlier.

Test-Style Questions

A test asks the same questions with more words. Read slowly. Circle the best answer.

9. A teacher recorded the number of minutes each of her 28 students spent on Khan Academy last night. The results are shown in the histogram in Problem 2. Which statement is supported by the histogram?

   A Exactly 8 students spent 20 minutes on Khan Academy.

   (B) The interval from 30 to 45 minutes contains more students than any other interval.

   C No student spent more than 60 minutes on Khan Academy.

   D Half of the students spent less than 30 minutes on Khan Academy.

10. The distribution of a data set is skewed to the right. Which statement best describes the relationship between the mean and the median of the data set?

    A The mean is less than the median.

    (B) The mean is greater than the median.

    C The mean is equal to the median.

    D The median is twice the mean.

11. The five-number summary of a data set is 12, 18, 25, 31, 40. What is the interquartile range (IQR) of the data set?

    A 7

    (B) 13

    C 25

    D 28

12. The box plots in Problem 7 show the lengths of fish caught in Lake A and Lake B. Which statement is true?

    A The median length in Lake A is greater than the median length in Lake B.

    (B) The interquartile range of Lake A is less than the interquartile range of Lake B.

    C The lengths in Lake A have a greater range than the lengths in Lake B.

    D About 50% of the fish in Lake B are longer than 19 inches.

13. For a data set, the first quartile is 40 and the third quartile is 52. Using the $1.5 \times$ IQR rule, which value would be an outlier?

    A 25

    B 35

    C 68

    (D) 74

IQR = $52 - 40 = 12$. $1.5 \times 12 = 18$. Fences: $40 - 18 = 22$ and $52 + 18 = 70$. 74 is above 70.